

Huygens Principle (Slit)

Secondary sources on slit; superposition & diffraction sketch; not full optics.

Pause

Reset phase

Incident plane wave (left) · Secondary sources on slit · Superposition (right)
Narrower slit → wider diffraction spread (qualitative)

WAVE & SLIT

Wavelength λ 0.22 (arb.)

Slit width a 1.8 (arb.)

secondary sources5

Animation f 1.6 Hz

OVERLAY

☒ Faint Huygens circles

Each point on the slit acts as a secondary source of circular wavelets. Their sum to the right reproduces a bent wavefront and spreading (Fraunhofer-style far-field is not computed — this is a near-field Huygens–Fresnel sketch).

SHORTCUTS

• Space / Enter — pause / resume

• R — reset phase

MEASURED VALUES	
$k = 2\pi/\lambda$	28.560
ω	10.053 rad/s
Sources on slit	5

About this model

Huygens' Principle provides a powerful geometric method for predicting how waves propagate. It states that every point on a wavefront can be considered a source of secondary spherical wavelets. The new wavefront at a later time is the envelope of these wavelets. This simulator applies this principle to the classic case of a plane wave incident on a single slit. It visualizes the construction of the diffracted wavefront by treating numerous point sources along the slit opening as emitters of circular wavelets. The user can observe how the superposition of these wavelets—their interference—creates the characteristic diffraction pattern on a screen. The model calculates the resulting intensity pattern using the principle of superposition and the analytic solution for single-slit Fraunhofer diffraction, where the intensity $I(\theta)$ is proportional to $[\sin(\beta)/\beta]^2$, with $\beta = (\pi a \sin\theta)/\lambda$, where 'a' is the slit width, λ is the wavelength, and θ is the angle from the central axis. Key simplifications include a 2D model (slit as a line source), monochromatic light, and the far-field (Fraunhofer) approximation where the screen is effectively at infinity. By interacting with the controls for slit width and wavelength, students learn how these parameters affect the diffraction pattern's width and spacing, directly connecting the mathematical model to a visual, physical process.

Who it's for: Undergraduate physics and engineering students studying wave optics, particularly those covering Huygens' Principle, superposition, and introductory diffraction theory.

KEY TERMS

- Huygens' Principle
- Diffraction
- Single Slit
- Wave Superposition
- Wavelets
- Fraunhofer Diffraction
- Interference
- Wavefront

How it works

Huygens pictured every point on a wavefront as a source of secondary wavelets; the **envelope** of those wavelets is the new front. A **slit** restricts where those sources exist, so the field spreads and **diffracts**. This page uses a discrete set of sources on the opening and **complex superposition** (cyan $\approx$ amplitude); it is not a full scalar diffraction integral.

Key equations

$$u \propto \sum_j \exp(i(k r_j - \omega t)), \text{ } r_j = \text{distance to slit element } j$$

Frequently asked questions

Why does the central bright spot get wider when the slit gets narrower? That seems backwards!

This is a key feature of wave diffraction. A narrower slit means the source of secondary wavelets is more confined, which increases the angular spread of the wavelets as they propagate. Mathematically, the angular width of the central maximum is proportional to λ/a . So, a smaller slit width 'a' results in a larger diffraction angle and a wider pattern on the screen.

Is Huygens' Principle just a mathematical trick, or do the secondary wavelets really exist?

Huygens' Principle is a conceptual model and a powerful calculation tool. The secondary wavelets are not physically distinct sources; they are a construction that correctly predicts how a wavefront evolves. Its success, especially when modified by Fresnel to include interference, confirms it as a valid description of wave propagation, even though it doesn't describe the underlying mechanism of the wave medium itself.

This simulator shows a 2D slice. How does it relate to a real, 3D single-slit experiment?

The simulator shows a cross-section. A real slit is a long, narrow rectangle. The 2D model accurately represents the diffraction pattern in the dimension where the slit is narrow (width 'a'). In the other dimension (along the slit's length), the slit is very wide, so little diffraction occurs there, resulting in a pattern of elongated stripes parallel to the slit, which is what this 2D intensity profile would produce if extended.

What is the main limitation of the model shown here?

The primary limitation is the Fraunhofer (far-field) approximation. It assumes the observing screen is very far from the slit, so that the waves arriving from different points in the slit are approximately parallel. For a screen close to the slit (the near-field or Fresnel regime), the wavefront curvature is significant, and the pattern is more complex, involving different interference conditions.

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Diffraction

▶ Launch Simulator

NEW

SCHOOL

Mach Cone (Schematic)

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FEATURED

SCHOOL

Wave Interference

▶ Launch Simulator

Single and double slit with interference patterns.

$M = v/c > 1$: Huygens pulses + cone $\sin \mu = 1/M$; not CFD shock.

Two sources creating constructive and destructive patterns. 2D ripple tank.

SCHOOL

Beat Frequency

▶ Launch Simulator

Two slightly different frequencies creating audible beats.

NEW

SCHOOL

Active Noise Cancellation (1D)

▶ Launch Simulator

Two tones same f : amplitudes and phase; RMS sum vs ideal π phase. Destructive interference demo.

NEW

SCHOOL

Acoustic Phased Array (Line)

▶ Launch Simulator

N sources, spacing d/λ and phase ramp: steer audio/sonar-like beams (array factor sketch).